SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I V. Sheershika (192492056) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student V. Sheershika

Assessment 1: TCP Three-Way Handshake in Wireshark

Aim

To capture TCP packets during connection setup and identify the **SYN, SYN-ACK, and ACK** packets.

Procedure

1. Open Wireshark and select the active **Wi-Fi** interface.
2. Start packet capture.
3. Open Command Prompt and execute:

   curl https://example.com

4. Stop the capture.
5. Apply the filter:

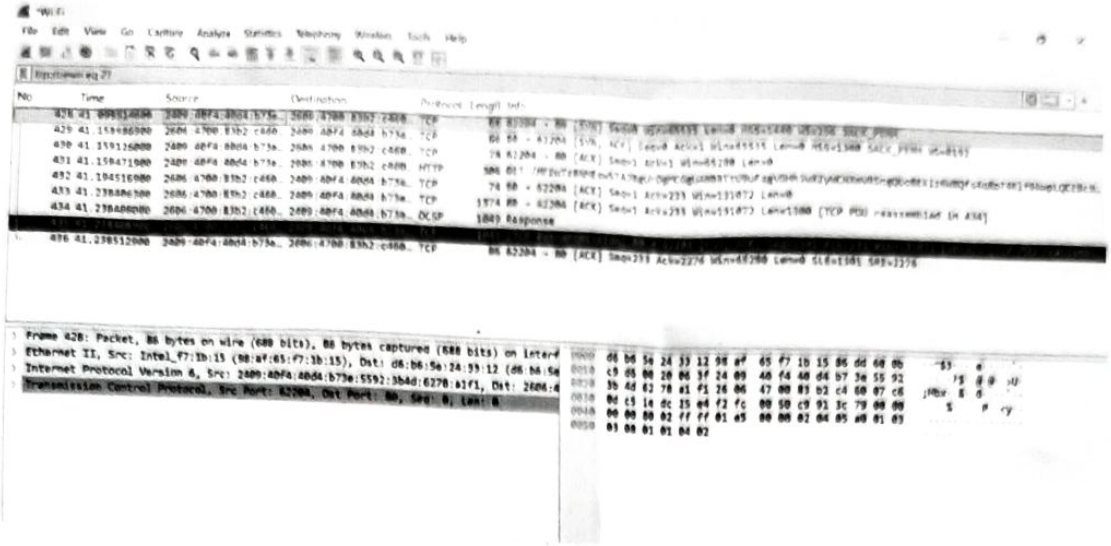
   tcp.flags.syn == 1

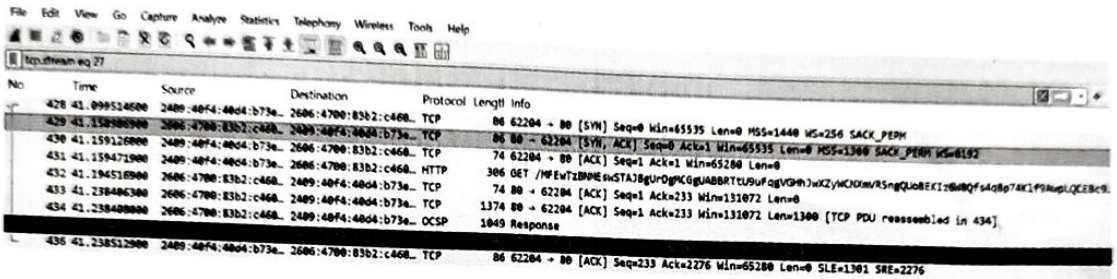
6. Identify the **SYN** and **SYN-ACK** packets.
7. Select the corresponding TCP conversation to identify the final **ACK** packet.
8. Observe the TCP flags in the packet details.

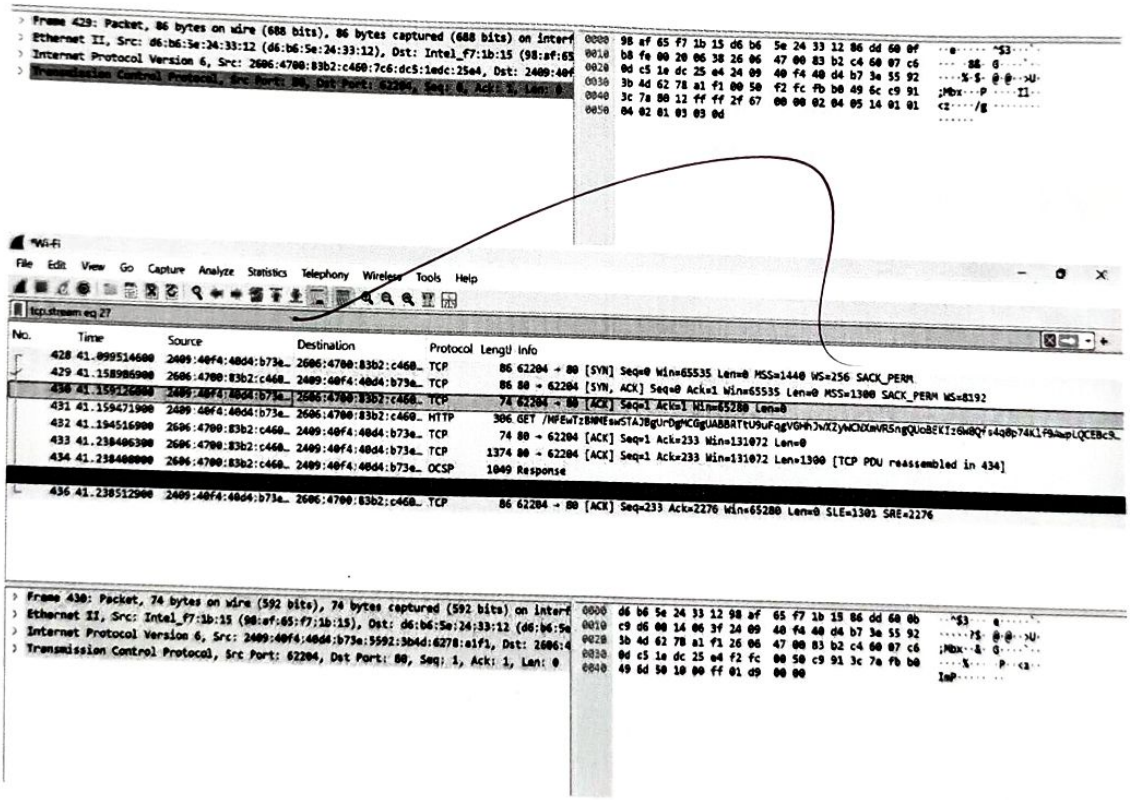
Output / Observation

Packet	TCP Flag	Purpose
1	**SYN**	Client requests a TCP connection
2	**SYN, ACK**	Server accepts the request and sends its own SYN
3	**ACK**	Client acknowledges the server

Result







Assessment 2: DNS Query Using UDP in Wireshark

Aim

To capture a DNS query using UDP in Wireshark and compare the UDP header with the TCP header.

Procedure

1. Open Wireshark and select Wi-Fi.
2. Start packet capture.
3. Open Command Prompt.
4. Enter:

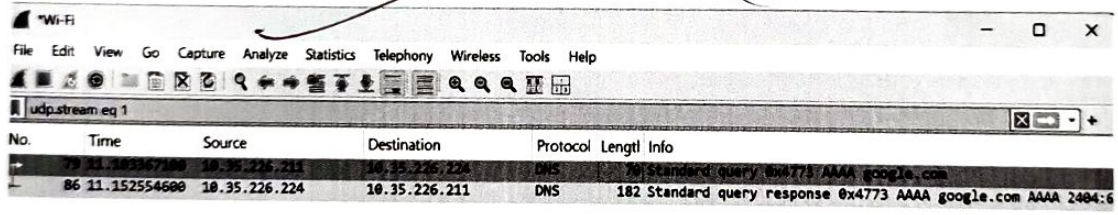
   nslookup google.com

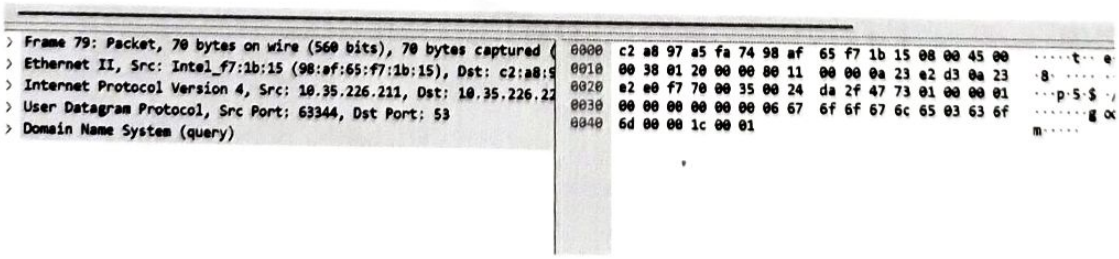
5. Stop the capture in Wireshark.
6. Apply the filter:

   dns && udp

7. Select the DNS query packet.
8. Expand User Datagram Protocol and observe the UDP header.

Output:





Assessment 3: TCP Retransmission vs UDP

Aim

To simulate packet loss and observe how TCP retransmits lost packets, while UDP does not retransmit them.

Procedure

Part A — Capture TCP retransmission

1. Open Wireshark.
2. Select Wi-Fi and start capturing.
3. Open Command Prompt.
4. Generate TCP traffic:

curl https://example.com

5. While traffic is being generated, briefly turn Wi-Fi OFF and ON.
6. Generate the traffic again if necessary.
7. Stop Wireshark.
8. Apply this filter:

tcp.analysis.retransmission

Wireshark may display:

[TCP Retransmission]

This indicates that TCP retransmitted a packet.

Part B — Observe UDP

Generate DNS traffic:

nslookup google.com

Use the filter:

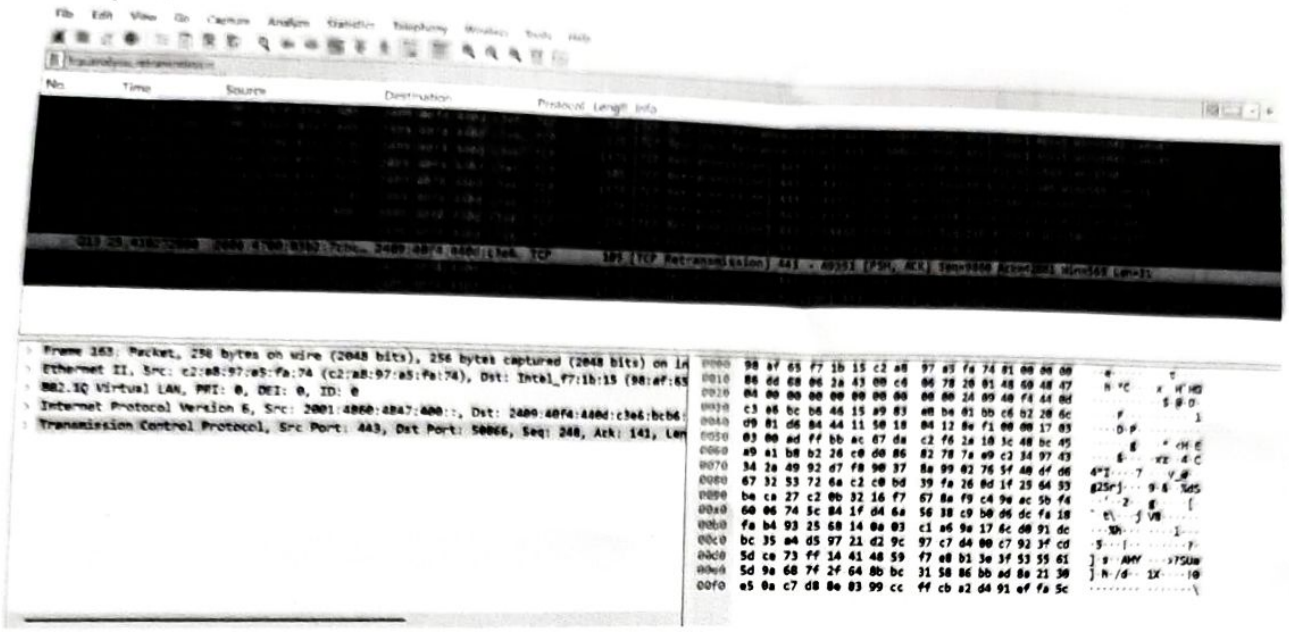
dns && udp

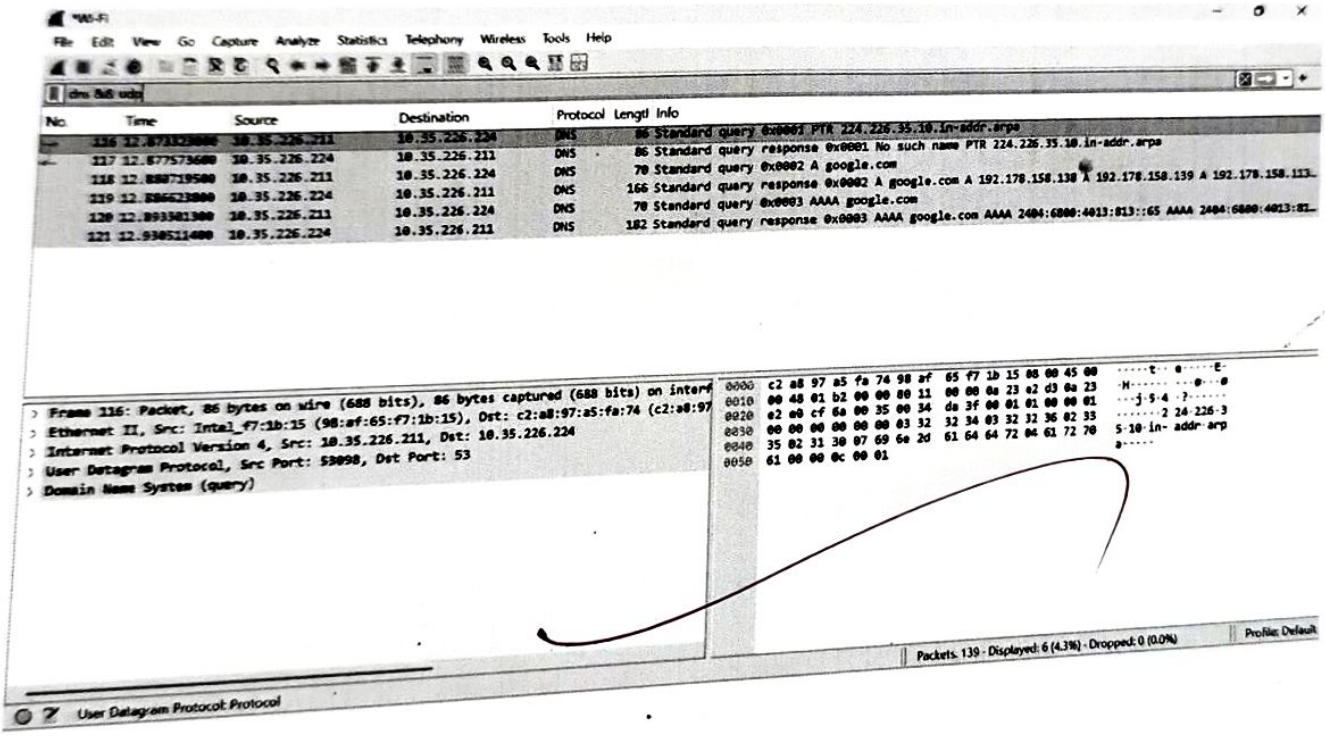
You will see:

DNS Query
DNS Response

UDP itself does not show a TCP-style retransmission when a packet is lost.

Output:





4: DNS Traffic and Query Types in Wireshark

Aim

To capture DNS traffic, identify A, AAAA and MX query types, observe the response, and measure DNS resolution time.

Procedure

1. Open Wireshark.
2. Select Wi-Fi and click Start Capture.
3. Open Command Prompt.
4. Run:

```
nslookup google.com
```

5. Stop the capture.
6. Apply the filter:

```
dns
```

Step 1: Identify A Record

In CMD:

```
nslookup -type=A google.com
```

In Wireshark, look for:

Standard query A google.com

The response will contain an IPv4 address, for example:

google.com → 142.x.x.x

A = IPv4 address

Step 2: Identify AAAA Record

Run:

```
nslookup -type=AAAA google.com
```

Wireshark should show:

Standard query AAAA google.com
Standard query response AAAA google.com

AAAA = IPv6 address

Your previous screenshot already showed this successfully:

Standard query 0x0003 AAAA google.com

Standard query response 0x0003 AAAA google.com
So you already have a good AAAA output.

Step 3: Identify MX Record

Run:

nslookup -type=MX google.com

Wireshark should show:

Standard query MX google.com
Standard query response MX google.com

MX = Mail Exchange record, which identifies mail servers for the domain.

Step 4: Measure Resolution Time

In Wireshark, find:

DNS Query

and its matching:

DNS Response

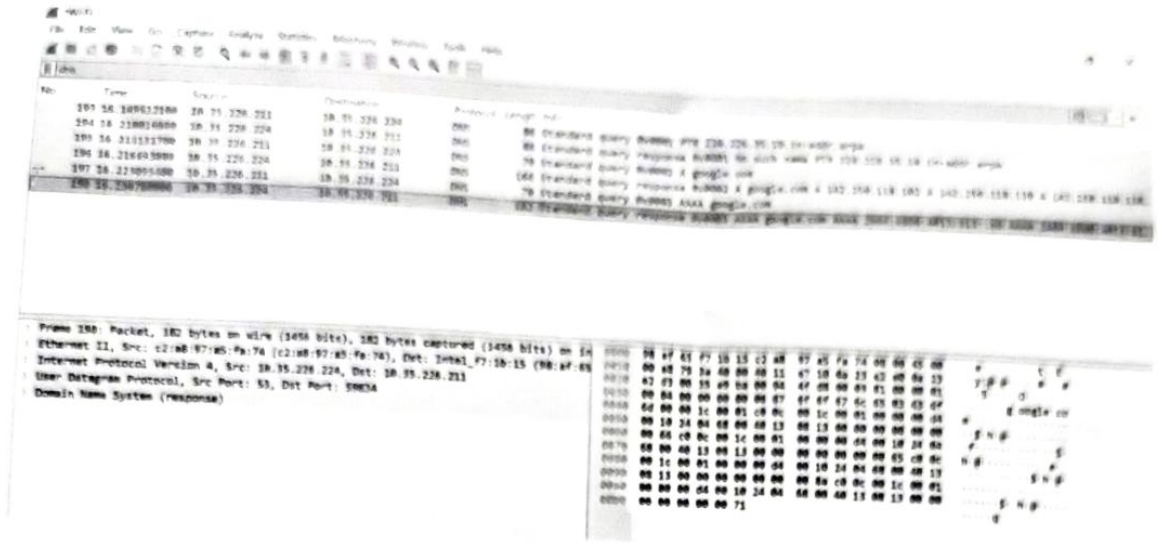
For example:

Query 12.893301 sec
Response 12.930511 sec

Therefore:

Resolution time = 12.930511 - 12.893301
= 0.037210 seconds
= 37.21 ms

Your actual time will be different.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation